

Exact Polynomial Windows from Finite Sums of Shifted and Scaled Rvachev u Functions

A constructive arbitrary-interval theory with binary partitions, Appell transforms, and support–sparsity tradeoffs

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Abstract

Let $u = u_p$ be the compactly supported C^∞ Rvachev function, normalized by $\text{supp } u = [-1, 1]$ and $\int u = 1$. This report gives a constructive answer to the following local exact-representation problem: given a polynomial p of degree at most d and a nondegenerate interval $[a, b]$, represent p on the entire closed interval as a *finite* sum of shifted and scaled copies of u .

The central identity is an explicit locally finite formula for every repeated left antiderivative of a scaled u atom. Its coefficients are restricted binary partition numbers

$$p_m(N) = [z^N] \prod_{j=0}^{m-1} (1 - z^{2^j})^{-1}.$$

A support truncation converts that half-line identity into an exact finite formula on $[a, b]$. For an arbitrary scale parameter $h > 0$, the construction uses $(d+1)(1 + \lceil (b-a)/(2h) \rceil)$ atoms. In the sparse choice $h = (b-a)/2$, every basis polynomial uses exactly two atoms; hence every $p \in \mathcal{P}_d$ has a canonical representation with at most $2d+2$ raw u atoms. The coefficients are computed exactly by applying the reciprocal moment-generating operator $M(D)^{-1}$, whose exponential coefficients are complete Bell polynomials in Bernoulli-number cumulants of the u distribution.

Further results include a monic Appell basis, a closed Wronskian and interpolation determinant, an exact compactly supported C^∞ extension operator, a support–atom Pareto law, a Thue–Morse Green-function interpretation of the binary weights, rationality and covariance properties, and numerical conditioning analysis. The report explicitly separates classical prior existence results of V. L. and V. A. Rvachev from formulas that appear absent from the reviewed **ProveIt** corpus. Worldwide priority is not asserted for those formulas. A fully commented Python implementation performs exact symbolic construction and an independent sinc-product Fourier check.

Status convention. **classical** marks material already present in the literature or the repository; **proved here** marks statements proved in this report; **corpus-new** means that the statement was not found in the recursively reviewed LaTeX corpus, not that worldwide novelty has been established; **conjectural** marks unproved proposals.

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1 Executive summary

The main conclusions are as follows.

1. **Exactness is possible only locally, but it is completely constructive.** A nonzero polynomial cannot equal a finite compactly supported sum on all of $\mathbb{R}$. On any prescribed compact interval, however, an exact identity exists.
2. **One antiderivative generates a dyadic train.** If

$$(Jf)(x) = \int_{-\infty}^x f(t) dt,$$

then

$$Ju(x) = \sum_{k \geq 0} u\left(\frac{x - (2k + 1)}{2}\right),$$

with a locally finite sum. Iterating this identity produces restricted binary partition numbers.

3. **Repeated antiderivatives have polynomial tails.** If Z has density u and

$$A_n(y) = \mathbb{E}(y - Z)^n,$$

then A_n is a monic Appell polynomial and, beyond the support of an atom, J^{n+1} of that atom equals a constant multiple of A_n .

4. **Finite truncation is exact on an interval.** With $L = b - a$, $h > 0$, and $K = \lceil L/(2h) \rceil$, define

$$\Psi_{n,h}^{[a,b]}(x) = n!2^{n(n+1)/2} \sum_{N=0}^K p_{n+1}(N) u\left(\frac{x - a - h(2^{n+1} - 2 + 2N)}{2^{n+1}h}\right).$$

Then, for every $x \in [a, b]$,

$$\Psi_{n,h}^{[a,b]}(x) = A_n\left(\frac{x - a + h}{h}\right).$$

Consequently $\{\Psi_{0,h}, \dots, \Psi_{d,h}\}$ is a basis of $\mathcal{P}_d$ on the interval.

5. **Two atoms per degree suffice.** For $h = L/2$ one has $K = 1$ and $p_m(0) = p_m(1) = 1$. Thus

$$\Psi_n^{[a,b]}(x) = n!2^{n(n+1)/2} \left[u\left(\frac{x - a - (2^n - 1)L}{2^n L}\right) + u\left(\frac{x - a - 2^n L}{2^n L}\right) \right]$$

equals $A_n(1 + 2(x - a)/L)$ on $[a, b]$. A degree- d polynomial therefore uses at most $2d + 2$ atoms.

6. **The coefficients are explicit.** Let $c = a - h$, $\tilde{p}(y) = p(c + hy)$, and

$$M(t) = \mathbb{E}e^{tZ} = \prod_{j \geq 1} \frac{\sinh(t/2^j)}{t/2^j}, \quad \frac{1}{M(t)} = \sum_{r \geq 0} \gamma_r \frac{t^r}{r!}.$$

Then

$$p(x) = \sum_{n=0}^d b_n \Psi_{n,h}^{[a,b]}(x), \quad b_n = \sum_{r=0}^{d-n} \frac{\gamma_r h^{n+r}}{n! r!} p^{(n+r)}(c).$$

All quantities are rational when the input data are rational.

7. **Support can be traded for atom count.** The construction is supported in

$$[a - 2h, a + h(2^{d+2} - 2 + 2K)].$$

Smaller h gives a tighter collar but more atoms. In particular, any requested right collar $\rho > 0$ is achieved by $h = \rho/2^{d+2}$ using at most

$$(d+1) \left(1 + \left\lceil \frac{2^{d+1}L}{\rho} \right\rceil \right)$$

atoms.

8. **Thue–Morse enters as a discrete inverse.** The finite Thue–Morse polynomial

$$T_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}) = \sum_{r=0}^{2^m-1} (-1)^{s_2(r)} z^r$$

has $\{p_m(N)\}$ as its causal convolution inverse. Thus the representation weights are Green coefficients for a Thue–Morse finite-difference operator.

2 Scope, corpus audit, and prior art

2.1 Recursive repository audit

The requested source tree was

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>.

The recursive audit used the repository’s own “second wave” inventory as a path-completeness check. At the reviewed revision that inventory counted 67 LaTeX paths, grouped as shown in Table 1.

Table 1: Organization of the recursively reviewed LaTeX corpus.

Class	Count	Role in the audit
Living syntheses and references	5	Current definitions, notation, global identities, glossary, and audit metadata.
Frozen historical documents	32	Earlier reports and derivations, used to detect rediscovery and changes of notation.
Active frontier drafts	23	Representation, inverse, integration, transform, q -series, Thue–Morse, and Fourier-decay investigations; 12 are Fourier-decay dossiers.
Transcribed or corrected papers	7	Literature-facing source material and historical comparison.
Total	67	All paths were included in the recursive search and thematic comparison.

The closest internal sources were the representation syntheses `Representation_Frontiers.tex` and `Representation_Second_Wave.tex`; the repeated-integration and integral-transform drafts; the Appell/Bell-polynomial material; the finite and infinite sinc-product studies; and the Thue–Morse/ q -series files. The corpus already contains, among much else,

- the relation between the Fabius and Rvachev functions;
- the refinement differential equation and infinite box-convolution model;
- the probability law, moments, cumulants, and Bell-polynomial formulas;
- Fourier products and the integer zero multiplicities $1 + v_2(m)$;
- dyadic Taylor/Appell blocks for the Fabius function;
- repeated integration as a route from finite box splines to u ;
- many representation formulas for u , the Fabius function, and their transforms.

No formula equivalent to the interval basis in [Theorem 6.1](#), the binary-partition antiderivative identity in [Theorem 4.3](#), or the coefficient transform in [Theorem 7.1](#) was found in the reviewed corpus. This is the meaning of [corpus-new](#) below.

2.2 Classical polynomial-representation results

It would be incorrect to claim that polynomial representation by atomic functions is new in itself. The 1990 survey of V. A. Rvachev states that the spaces denoted there by UP_n contain all polynomials of degree at most n and cites the 1972 paper of V. L. Rvachev and V. A. Rvachev, “On the representation of polynomials by functions with compact support” [\[6, 8\]](#). The survey also develops generalized Taylor systems made from finite-function shifts.

The contribution attempted here is narrower and more explicit:

- a closed arbitrary-interval formula expressed only through shifted and scaled copies of the single function u ;
- a locally finite repeated-antiderivative identity with coefficients $p_m(N)$;
- an immediately implementable exact coefficient transform;
- the linear upper bound $2d + 2$ in a canonical sparse construction;
- exact support bounds and a tunable collar-atom tradeoff;
- a Thue–Morse Green-function interpretation;
- closed Wronskian and interpolation determinants;
- a reproducible symbolic and numerical algorithm.

These formulas are presented as [proved here](#) and [corpus-new](#), while worldwide priority remains open pending a more exhaustive multilingual search of the atomic-function literature.

3 Normalization and basic facts

Definition 3.1 (Rvachev $u = \text{up}$). Throughout, $u : \mathbb{R} \rightarrow \mathbb{R}$ denotes the unique normalized compactly supported C^∞ function satisfying

$$\text{supp } u = [-1, 1], \quad u(x) = u(-x), \quad \int_{\mathbb{R}} u(x) \, dx = 1, \quad (1)$$

$$u'(x) = 2u(2x + 1) - 2u(2x - 1). \quad (2)$$

The associated Fabius function satisfies $F(x) = u(x - 1)$ on $[0, 1]$ and $u(x) = F(1 - |x|)$ for $|x| \leq 1$.

A useful probability realization is

$$Z = \sum_{j=1}^{\infty} 2^{-j} V_j, \quad V_j \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1], \quad (3)$$

with density u . Consequently

$$\hat{u}(\xi) = \mathbb{E} e^{-i\xi Z} = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{\xi}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z}, \quad (4)$$

$$M(t) = \mathbb{E} e^{tZ} = \prod_{j=1}^{\infty} \frac{\sinh(t/2^j)}{t/2^j}. \quad (5)$$

The variance is $\mu_2 = 1/9$. The first even moments are

$$\mu_0 = 1, \quad \mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}, \quad \mu_8 = \frac{132809}{32531625}. \quad (6)$$

Lemma 3.2 (Strict one-sided monotonicity). *The function u is strictly increasing on $(-1, 0)$ and strictly decreasing on $(0, 1)$.*

Proof. For $-1 < x < 0$, the term $u(2x - 1)$ vanishes while $2x + 1 \in (-1, 1)$ and $u(2x + 1) > 0$, so (2) gives $u'(x) > 0$. The case $0 < x < 1$ follows symmetrically. Positivity in the open support follows, for example, from the infinite convolution model (3). $\square$

In particular, one atom cannot equal a nonzero constant on a nondegenerate interval. At least two atoms are required even in degree zero.

4 The antiderivative train

Let

$$(Jf)(x) = \int_{-\infty}^x f(t) dt. \quad (7)$$

The following elementary identity drives the entire construction.

Proposition 4.1 (One-step antiderivative identity [proved here](#), [corpus-new](#)). *For every $x \in \mathbb{R}$,*

$$Ju(x) = \sum_{k=0}^{\infty} u\left(\frac{x - (2k + 1)}{2}\right), \quad (8)$$

where the sum is locally finite.

Proof. The k th summand is supported on $[2k - 1, 2k + 3]$, so only finitely many terms are nonzero at a fixed x . Differentiating and using (2),

$$\begin{aligned} \frac{d}{dx} u\left(\frac{x - (2k + 1)}{2}\right) &= \frac{1}{2} u'\left(\frac{x - (2k + 1)}{2}\right) \\ &= u(x - 2k) - u(x - 2k - 2). \end{aligned}$$

The locally finite series telescopes to $u(x)$. Both sides vanish for sufficiently negative x , so the integration constant is zero. $\square$

For a scale $h > 0$ and center c , write

$$u_{h,c}(x) = u\left(\frac{x - c}{h}\right). \quad (9)$$

Then [Theorem 4.1](#) becomes

$$Ju_{h,c}(x) = h \sum_{k \geq 0} u\left(\frac{x - c - h(2k + 1)}{2h}\right). \quad (10)$$

4.1 Restricted binary partitions

Definition 4.2. For $m \geq 1$, let $p_m(N)$ be the restricted binary partition number

$$p_m(N) = \# \left\{ (k_0, \dots, k_{m-1}) \in \mathbb{N}^m : N = \sum_{j=0}^{m-1} 2^j k_j \right\}. \quad (11)$$

Equivalently,

$$\sum_{N \geq 0} p_m(N) z^N = \prod_{j=0}^{m-1} \frac{1}{1 - z^{2^j}}. \quad (12)$$

The first cases are

$$p_1(N) = 1, \quad p_2(N) = \left\lfloor \frac{N}{2} \right\rfloor + 1,$$

and the recurrence

$$p_m(N) = p_{m-1}(N) + p_m(N - 2^{m-1}) \quad (13)$$

holds with $p_m(N) = 0$ for $N < 0$.

Theorem 4.3 (Repeated antiderivative formula [proved here](#), [corpus-new](#)). *For every integer $m \geq 1$,*

$$J^m u_{h,c}(x) = h^m 2^{m(m-1)/2} \sum_{N=0}^{\infty} p_m(N) u\left(\frac{x - c - h(2^m - 1 + 2N)}{2^m h}\right). \quad (14)$$

The sum is locally finite.

Proof. The case $m = 1$ is (10). Suppose the formula holds at m . Integrate each locally finite atom by (10). An atom with index N has scale $2^m h$ and center $c + h(2^m - 1 + 2N)$, so a new index $k \geq 0$ contributes scale $2^{m+1} h$ and center

$$\begin{aligned} c + h(2^m - 1 + 2N) + 2^m h(2k + 1) \\ = c + h(2^{m+1} - 1 + 2(N + 2^m k)). \end{aligned}$$

The prefactor is multiplied by $2^m h$, becoming $h^{m+1} 2^{m(m+1)/2}$. The multiplicity of the new index N' is

$$\sum_{k \geq 0} p_m(N' - 2^m k) = p_{m+1}(N'),$$

which is exactly (13). This proves the induction.

For local finiteness, the atom indexed by N has left support endpoint computed in [Theorem 4.4](#); this endpoint tends to $+\infty$ with N . $\square$

Corollary 4.4 (Support-edge invariance [proved here](#), [corpus-new](#)). *The N th atom on the right side of (14) has support*

$$\left[c - h + 2hN, c + h(2^{m+1} - 1 + 2N) \right]. \quad (15)$$

In particular, its left endpoint $c - h + 2hN$ is independent of the antiderivative order m .

This invariance is the geometric reason a single truncation index can be used simultaneously for every polynomial degree.

4.2 Thue–Morse as the inverse difference operator

Let $s_2(r)$ denote the binary digit sum and $\varepsilon_r = (-1)^{s_2(r)}$. The finite Prouhet–Thue–Morse polynomial is

$$T_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}) = \sum_{r=0}^{2^m-1} \varepsilon_r z^r. \quad (16)$$

Proposition 4.5 (Thue–Morse Green sequence [proved here](#), [corpus-new](#)). *The restricted binary partition sequence is the causal convolution inverse of the length- 2^m signed Thue–Morse block:*

$$\sum_{r=0}^{\min(N, 2^m-1)} (-1)^{s_2(r)} p_m(N-r) = \delta_{N0}. \quad (17)$$

Equivalently, if S is the backward shift on sequences, then

$$\left[\prod_{j=0}^{m-1} (I - S^{2^j}) \right] p_m = \delta_0. \quad (18)$$

Proof. Multiply the generating functions (12) and (16). Their product is one. Coefficient extraction gives (17), and the shift-operator form is the same convolution identity. $\square$

Thus the continuous inverse of the refinement derivative and the discrete inverse of a Thue–Morse difference stencil appear in the same formula. This is more than a decorative analogy: the weights in the exact polynomial window are the Green coefficients that undo all m dyadic differences.

Proposition 4.6 (Quasipolynomial structure [proved here](#)). *For fixed m , $p_m(N)$ is a quasipolynomial in N of degree $m-1$ and period dividing 2^{m-1} . Its leading term is independent of the residue class:*

$$p_m(N) = \frac{N^{m-1}}{(m-1)! 2^{m(m-1)/2}} + \mathcal{O}(N^{m-2}). \quad (19)$$

Consequently, for $m = n+1$ the normalized weight

$$w_{n,N} = n! 2^{n(n+1)/2} p_{n+1}(N) \quad (20)$$

is a monic degree- n quasipolynomial: $w_{n,N} = N^n + \mathcal{O}(N^{n-1})$.

Proof. The generating function (12) is rational and all its poles are 2^{m-1} th roots of unity. Partial fractions therefore give a finite sum of polynomial multiples of root-of-unity exponentials, hence a quasipolynomial with period dividing 2^{m-1} . The pole at $z = 1$ has order m . Since

$$1 - z^{2^j} = 2^j(1 - z) + \mathcal{O}((1 - z)^2),$$

we have

$$\prod_{j=0}^{m-1} (1 - z^{2^j}) = 2^{m(m-1)/2} (1 - z)^m (1 + \mathcal{O}(1 - z)).$$

The coefficient of $(1 - z)^{-m}$ is $\binom{N+m-1}{m-1}$, giving (19). $\square$

5 Polynomial tails and the u -Appell family

Definition 5.1. Let Z have density u . Define

$$A_n(y) = \mathbb{E}(y - Z)^n. \quad (21)$$

By symmetry of Z , this is also $\mathbb{E}(y + Z)^n$.

Proposition 5.2 (Appell structure [classical](#)/[proved here](#)). *The polynomials A_n are monic and satisfy*

$$\sum_{n=0}^{\infty} A_n(y) \frac{t^n}{n!} = e^{yt} M(t), \quad (22)$$

$$A'_n(y) = n A_{n-1}(y), \quad (23)$$

$$A_n(y) = \sum_{r=0}^n \binom{n}{r} \mu_r y^{n-r}. \quad (24)$$

Proof. Take expectations in $e^{t(y-Z)}$. Symmetry gives $\mathbb{E}e^{-tZ} = M(t)$. Expanding the exponential gives (22); differentiation and coefficient extraction give the other identities. $\square$

The first members are

$$\begin{aligned} A_0(y) &= 1, \\ A_1(y) &= y, \\ A_2(y) &= y^2 + \frac{1}{9}, \\ A_3(y) &= y^3 + \frac{1}{3}y, \\ A_4(y) &= y^4 + \frac{2}{3}y^2 + \frac{19}{675}, \\ A_5(y) &= y^5 + \frac{10}{9}y^3 + \frac{19}{135}y, \\ A_6(y) &= y^6 + \frac{5}{3}y^4 + \frac{19}{45}y^2 + \frac{583}{59535}. \end{aligned}$$

Theorem 5.3 (Tail-polynomial identity [proved here](#)). *For $n \geq 0$ and $x \geq c + h$,*

$$J^{n+1}u_{h,c}(x) = \frac{h^{n+1}}{n!} A_n\left(\frac{x-c}{h}\right). \quad (25)$$

Proof. Repeated integration gives the standard truncated-power kernel

$$J^{n+1}u_{h,c}(x) = \frac{1}{n!} \int_{-\infty}^x (x-t)^n u\left(\frac{t-c}{h}\right) dt.$$

For $x \geq c + h$, the full support $[c-h, c+h]$ lies below x . Substitute $t = c + hz$ to obtain

$$\frac{h^{n+1}}{n!} \int_{-1}^1 \left(\frac{x-c}{h} - z\right)^n u(z) dz,$$

which is (25). $\square$

Combining [Theorems 4.3](#) and [5.3](#) yields a locally finite half-line representation that is already of independent interest.

Corollary 5.4 (Half-line polynomial identity [proved here](#), [corpus-new](#)). *For $x \geq c + h$,*

$$A_n\left(\frac{x-c}{h}\right) = n! 2^{n(n+1)/2} \sum_{N \geq 0} p_{n+1}(N) u\left(\frac{x-c-h(2^{n+1}-1+2N)}{2^{n+1}h}\right). \quad (26)$$

The right side is locally finite.

6 The exact finite-window theorem

We now truncate the half-line train using the support-edge invariance.

Theorem 6.1 (Exact finite polynomial window [proved here](#), [corpus-new](#)). *Let $a < b$, $h > 0$, $L = b - a$, and*

$$K = \left\lceil \frac{L}{2h} \right\rceil. \quad (27)$$

For every $n \geq 0$, define

$$\boxed{\Psi_{n,h}^{[a,b]}(x) = n!2^{n(n+1)/2} \sum_{N=0}^K p_{n+1}(N) \times u\left(\frac{x - a - h(2^{n+1} - 2 + 2N)}{2^{n+1}h}\right)}. \quad (28)$$

Then on the full closed interval $[a, b]$,

$$\Psi_{n,h}^{[a,b]}(x) = A_n\left(\frac{x - a + h}{h}\right). \quad (29)$$

Consequently, for every $d \geq 0$,

$$\text{span}\{\Psi_{0,h}^{[a,b]}, \dots, \Psi_{d,h}^{[a,b]}\}_{|[a,b]} = \mathcal{P}_d. \quad (30)$$

Proof. Set $c = a - h$. Then $x \geq a = c + h$ throughout the interval, so [Theorem 5.4](#) applies and its atom centers become

$$c + h(2^{n+1} - 1 + 2N) = a + h(2^{n+1} - 2 + 2N).$$

By [Theorem 4.4](#), the left endpoint of the atom indexed by N is

$$c - h + 2hN = a - 2h + 2hN.$$

For every omitted index $N \geq K + 1$, this endpoint satisfies

$$a - 2h + 2hN \geq a + 2hK \geq b.$$

Thus every omitted atom vanishes on $[a, b]$; at a possible equality $x = b$, its argument is -1 and $u(-1) = 0$. The finite truncation therefore agrees exactly with the full half-line identity, proving (29). Since A_n is monic of degree n , the displayed family is a polynomial basis. $\square$

Remark 6.2 (Why the interval is essential). Every finite sum in (28) is compactly supported. A nonzero polynomial is not compactly supported, so a global identity on all of $\mathbb{R}$ is impossible. The theorem constructs an exact *window*: polynomial on the specified interval, smoothly transitioning elsewhere, and identically zero outside a finite support.

6.1 Two atoms per degree

Corollary 6.3 (Sparse two-atom basis [proved here](#), [corpus-new](#)). *If $h \geq L/2$, then $K = 1$ and*

$$\Psi_{n,h}^{[a,b]}(x) = n!2^{n(n+1)/2} \left[u\left(\frac{x - a - h(2^{n+1} - 2)}{2^{n+1}h}\right) + u\left(\frac{x - a - h2^{n+1}}{2^{n+1}h}\right) \right]. \quad (31)$$

It equals $A_n((x - a + h)/h)$ on $[a, b]$.

Proof. Since $L/(2h) \in (0, 1]$, its ceiling is one. Also $p_m(0) = p_m(1) = 1$ for all $m \geq 1$. $\square$

The smallest support within this sparse causal family is obtained at $h = L/2$.

Corollary 6.4 (Canonical interval formula [proved here](#), [corpus-new](#)). *Let $L = b - a$. Define*

$$\boxed{\Psi_n^{[a,b]}(x) = n!2^{n(n+1)/2} \left[u\left(\frac{x-a-(2^n-1)L}{2^n L}\right) + u\left(\frac{x-a-2^n L}{2^n L}\right) \right]}. \quad (32)$$

Then

$$\Psi_n^{[a,b]}(x) = A_n \left(1 + \frac{2(x-a)}{L} \right), \quad a \leq x \leq b. \quad (33)$$

Every polynomial of degree at most d has a representation using at most $2d + 2$ raw u atoms.

For $n = 0$, (32) reduces to the familiar partition identity

$$u\left(\frac{x-a}{L}\right) + u\left(\frac{x-b}{L}\right) = 1, \quad a \leq x \leq b. \quad (34)$$

The theorem can therefore be viewed as a complete higher-degree lift of the basic two-shift partition of unity.

7 Exact coefficient extraction

The basis theorem becomes an algorithm once the polynomial is converted to the Appell coordinates.

7.1 Cumulants, Bernoulli numbers, and the reciprocal MGF

Write

$$\log M(t) = \sum_{r \geq 1} \kappa_r \frac{t^r}{r!}. \quad (35)$$

From the product (5) and the classical expansion of $\log(\sinh z/z)$,

$$\kappa_{2m} = \frac{2^{2m-1} B_{2m}}{m(2^{2m} - 1)}, \quad \kappa_{2m+1} = 0. \quad (36)$$

Define

$$G(t) = \frac{1}{M(t)} = \sum_{r=0}^{\infty} \gamma_r \frac{t^r}{r!}. \quad (37)$$

Then

$$\gamma_r = Y_r(-\kappa_1, \dots, -\kappa_r), \quad (38)$$

where Y_r is the complete exponential Bell polynomial. Equivalently,

$$\gamma_0 = 1, \quad \gamma_n = - \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \gamma_{n-r}. \quad (39)$$

The first values are

$$\begin{aligned} \gamma_0 &= 1, & \gamma_2 &= -\frac{1}{9}, & \gamma_4 &= \frac{31}{675}, \\ \gamma_6 &= -\frac{2347}{59535}, & \gamma_8 &= \frac{1904369}{32531625}, & \gamma_{10} &= -\frac{3309760579}{24405225075}, \end{aligned} \quad (40)$$

with all odd coefficients zero.

Theorem 7.1 (Coefficient transform [proved here](#), [corpus-new](#)). Let $p \in \mathcal{P}_d$, $c = a - h$, and

$$\tilde{p}(y) = p(c + hy). \quad (41)$$

There is a unique expansion

$$\tilde{p}(y) = \sum_{n=0}^d b_n A_n(y), \quad (42)$$

and its coefficients are given by any of the equivalent formulas

$$q(y) = G(D)\tilde{p}(y) = \sum_{r=0}^d \frac{\gamma_r}{r!} \tilde{p}^{(r)}(y), \quad b_n = [y^n]q(y), \quad (43)$$

$$b_n = \sum_{r=0}^{d-n} \frac{\gamma_r h^{n+r}}{n! r!} p^{(n+r)}(c), \quad (44)$$

$$b_n = \sum_{r=0}^{d-n} \binom{n+r}{r} \gamma_r c_{n+r} \quad \text{if} \quad \tilde{p}(y) = \sum_{j=0}^d c_j y^j. \quad (45)$$

Therefore

$$p(x) = \sum_{n=0}^d b_n \Psi_{n,h}^{[a,b]}(x), \quad x \in [a, b]. \quad (46)$$

Proof. The moment formula (24) says precisely that

$$A_n(y) = M(D)y^n. \quad (47)$$

Since $M(0) = 1$, the formal differential operator $M(D)$ is invertible on every finite-dimensional polynomial space, with inverse $G(D)$. Apply $G(D)$ to (42) to obtain $q(y) = \sum b_n y^n$, proving (43). Taking the coefficient of y^n , or equivalently the n th derivative at zero, gives (44). Direct coefficient extraction gives (45). Finally use (29). $\square$

The raw amplitude of the atom at level n and partition index N is therefore

$$\alpha_{n,N} = n! 2^{n(n+1)/2} b_n p_{n+1}(N). \quad (48)$$

In the sparse construction $N \in \{0, 1\}$ and both atoms at level n have the same amplitude

$$\alpha_n = n! 2^{n(n+1)/2} b_n. \quad (49)$$

7.2 A triangular alternative

If $\tilde{p}(y) = \sum c_n y^n$, monicity of A_n gives the descending recurrence

$$b_n = c_n - \sum_{k=n+1}^d b_k \binom{k}{n} \mu_{k-n}, \quad n = d, d-1, \dots, 0. \quad (50)$$

This is often the simplest exact-rational implementation when the moments are already available.

7.3 Closed monomial coefficients

Define the reciprocal Appell family

$$C_m(y) = G(D)y^m = \sum_{r=0}^m \binom{m}{r} \gamma_r y^{m-r}. \quad (51)$$

For $p(x) = (x - a)^m$, one has

$$b_n = h^m \binom{m}{n} C_{m-n}(-1), \quad 0 \leq n \leq m. \quad (52)$$

Thus all monomial windows admit a direct Bernoulli–Bell closed form with no linear solve.

8 Algorithm and arithmetic complexity

Algorithm 8.1 (Exact construction). Input a polynomial $p(x)$ of degree d , a nondegenerate interval $[a, b]$, and a scale $h > 0$.

1. Set $L = b - a$, $c = a - h$, and $K = \lceil L/(2h) \rceil$.
2. Compute $\kappa_0, \dots, \kappa_d$ from (36).
3. Compute $\gamma_0, \dots, \gamma_d$ by (39).
4. Form $\tilde{p}(y) = p(c + hy)$ and compute b_n using (44), (45), or (50).
5. Build the arrays $p_m(N)$, $1 \leq m \leq d + 1$, $0 \leq N \leq K$, by the dynamic recurrence (13).
6. For each $0 \leq n \leq d$ and $0 \leq N \leq K$, emit the atom

$$\begin{aligned} H_n &= 2^{n+1}h, \\ C_{n,N} &= a + h(2^{n+1} - 2 + 2N), \\ \alpha_{n,N} &= n! 2^{n(n+1)/2} b_n p_{n+1}(N). \end{aligned}$$

7. Return

$$\sum_{n=0}^d \sum_{N=0}^K \alpha_{n,N} u\left(\frac{x - C_{n,N}}{H_n}\right).$$

If the partition arrays are updated incrementally in m , the arithmetic operation count is

$$\mathcal{O}(d^2 + dK), \quad (53)$$

not counting coefficient bit growth. The output itself contains $(d + 1)(K + 1)$ atoms, so the dK term is unavoidable for an explicit atom list. In the sparse case $K = 1$, the output and construction are linear in d after the $\mathcal{O}(d^2)$ coefficient transform.

9 Structural properties

9.1 Uniqueness in the block basis

Proposition 9.1 (Block-basis uniqueness). *For fixed a, b, h, d , the coefficients $b_0, \dots, b_d$ in (46) are unique.*

Proof. The restrictions of $\Psi_{n,h}$ are the monic degree- n polynomials $A_n((x - a + h)/h)$. A triangular polynomial basis has unique coordinates. $\square$

This does *not* claim uniqueness if the amplitudes of all raw atoms are allowed to vary independently. The canonical object is the equal-amplitude binary-partition block at each degree.

9.2 Derivative ladder, Wronskian, and interpolation

Proposition 9.2 (Derivative ladder [proved here](#)). For $0 \leq k \leq n$ and $x \in [a, b]$,

$$\frac{d^k}{dx^k} \Psi_{n,h}^{[a,b]}(x) = \frac{(n)_k}{h^k} \Psi_{n-k,h}^{[a,b]}(x), \quad (54)$$

where $(n)_k = n(n-1) \cdots (n-k+1)$.

Proof. Differentiate (29) and use the Appell relation (23). $\square$

Theorem 9.3 (Wronskian and interpolation determinant [proved here](#), [corpus-new](#)). For every $x \in [a, b]$,

$$W(\Psi_{0,h}, \dots, \Psi_{d,h})(x) = h^{-d(d+1)/2} \prod_{n=0}^d n!. \quad (55)$$

For distinct nodes $x_0, \dots, x_d \in [a, b]$,

$$\det[\Psi_{j,h}(x_i)]_{i,j=0}^d = h^{-d(d+1)/2} \prod_{0 \leq i < j \leq d} (x_j - x_i). \quad (56)$$

Proof. After the affine change $y = (x - a + h)/h$, the basis consists of monic polynomials $A_j(y)$. Changing from monomials to a monic triangular basis has determinant one. The monomial Wronskian is $\prod j!$, and each derivative row contributes a factor h^{-k} . The interpolation determinant is the ordinary Vandermonde in the y_i variables. $\square$

Hence the block basis is an extended complete Chebyshev system on every subinterval. Polynomial interpolation in this basis is nonsingular at every set of distinct nodes.

9.3 Affine covariance and rationality

The entire construction is affine-covariant. If $x = \lambda X + \tau$ with $\lambda > 0$, centers and scales transform by the same affine map, while the Appell coefficient transform is exactly the usual polynomial pullback.

If $p \in \mathbb{Q}[x]$ and $a, b, h \in \mathbb{Q}$ with $h > 0$, then

- all moments, cumulants, and γ_r are rational;
- all b_n and amplitudes $\alpha_{n,N}$ are rational;
- every center and scale is rational, and all scale ratios are dyadic.

Thus the representation can be constructed without floating-point arithmetic.

9.4 A compactly supported smooth extension operator

For fixed d, a, b, h , define

$$E_{d,a,b,h}p(x) = \sum_{n=0}^d \sum_{N=0}^K \alpha_{n,N}(p) u\left(\frac{x - C_{n,N}}{H_n}\right). \quad (57)$$

Then $E_{d,a,b,h} : \mathcal{P}_d \rightarrow C_c^\infty(\mathbb{R})$ is linear and injective, and

$$(E_{d,a,b,h}p)(x) = p(x), \quad x \in [a, b]. \quad (58)$$

All derivatives match on the closed interval. Outside a finite support the extension vanishes identically, with infinite-order flatness at its external support endpoints inherited from u .

9.5 A norm bound

Since $0 \leq u \leq 1$, the global supremum norm satisfies

$$\|E_{d,a,b,h}p\|_\infty \leq \sum_{n=0}^d |b_n| n! 2^{n(n+1)/2} \sum_{N=0}^K p_{n+1}(N). \quad (59)$$

This bound is crude but useful as a symbolic stability certificate.

10 Support, sparsity, and localization

The support of the (n, N) atom in (28) is

$$[a - 2h + 2hN, a + h(2^{n+2} - 2 + 2N)]. \quad (60)$$

Therefore the full degree- d construction is supported in

$$\text{supp } E_{d,a,b,h}p \subseteq [a - 2h, a + h(2^{d+2} - 2 + 2K)]. \quad (61)$$

Because $2hK < L + 2h$,

$$\sup \text{supp } E_{d,a,b,h}p < b + 2^{d+2}h. \quad (62)$$

Theorem 10.1 (Collar-atom tradeoff [proved here](#), [corpus-new](#)). *Let $\rho > 0$ and choose*

$$h = \frac{\rho}{2^{d+2}}. \quad (63)$$

Then every $p \in \mathcal{P}_d$ has an exact u -atom representation on $[a, b]$ supported in $[a - \rho, b + \rho]$ and using at most

$$(d+1) \left(1 + \left\lceil \frac{2^{d+1}(b-a)}{\rho} \right\rceil \right) \quad (64)$$

atoms.

Proof. The left overhang is $2h = \rho/2^{d+1} \leq \rho$, and (62) gives a right overhang less than $2^{d+2}h = \rho$. The atom count is $(d+1)(K+1)$ with $K = \lceil (b-a)/(2h) \rceil$. $\square$

More generally, for asymmetric collars $\rho_-, \rho_+ > 0$, it is sufficient to choose

$$h \leq \min \left\{ \frac{\rho_-}{2}, \frac{\rho_+}{2^{d+2}} \right\}. \quad (65)$$

A reflected, right-causal construction interchanges the two collar roles.

In the sparse choice $h = L/2$, the exact support bound is

$$\text{supp } E_{d,a,b,L/2}p \subseteq [a - L, a + 2^{d+1}L]. \quad (66)$$

Thus linear atom count comes with an exponentially long one-sided scale ladder. The general- h theorem permits one to exchange that long tail for more atoms.

11 Relation to shift-invariant reproduction

A superficial reading might suggest a contradiction with the Strang–Fix conditions. There is none.

Under the Fourier convention (4), $\hat{u}$ has a zero at every nonzero $2\pi k$ of multiplicity

$$1 + v_2(k). \quad (67)$$

The minimum multiplicity is one, attained at odd k . Hence the single fixed-scale integer-shift space generated by u has only first-order global polynomial reproduction in the standard Strang–Fix sense: constants are the uniformly reproduced polynomials, but a fixed lattice of one scale does not globally reproduce arbitrary higher degrees [9, 10].

The present construction bypasses that obstruction in three simultaneous ways:

1. it asks for equality only on a bounded interval;
2. it is boundary-adapted and truncates atoms by support;
3. it uses a dyadic chain of scales $2^{n+1}h$ rather than one principal shift-invariant space.

This viewpoint helps reconcile the classical UP_n polynomial inclusion with modern Fourier reproduction theory. The exact window is a multiscale, boundary-generated object, not a global cardinal reproduction formula.

12 Worked exact example

Take

$$p(x) = 3x^4 - 2x^3 + 5x - 7, \quad [a, b] = [0, 1], \quad h = \frac{1}{2}. \quad (68)$$

Then $c = -1/2$ and

$$\tilde{p}(y) = p\left(\frac{y-1}{2}\right) = \frac{3}{16}y^4 - y^3 + \frac{15}{8}y^2 + y - \frac{145}{16}. \quad (69)$$

The exact Appell coefficients are

$$(b_0, b_1, b_2, b_3, b_4) = \left(-\frac{2084}{225}, \frac{4}{3}, \frac{7}{4}, -1, \frac{3}{16}\right). \quad (70)$$

After the normalization (49), the raw amplitudes are

$$(\alpha_0, \alpha_1, \alpha_2, \alpha_3, \alpha_4) = \left(-\frac{2084}{225}, \frac{8}{3}, 28, -384, 4608\right). \quad (71)$$

Thus, for every $0 \leq x \leq 1$,

$$p(x) = \sum_{n=0}^4 \alpha_n \left[u\left(\frac{x - (2^n - 1)}{2^n}\right) + u\left(\frac{x - 2^n}{2^n}\right) \right]. \quad (72)$$

The symbolic Appell residual is exactly zero.

The accompanying Python program also evaluates u independently by Fourier inversion of a 44-factor sinc product on a 2^{20} grid. On 1001 points of $[0, 1]$, the maximum numerical residual was

$$3.575315199456 \times 10^{-8}, \quad (73)$$

and the RMS residual was $1.211561786651 \times 10^{-8}$. These errors measure the numerical Fourier grid and interpolation, not the exact symbolic identity.

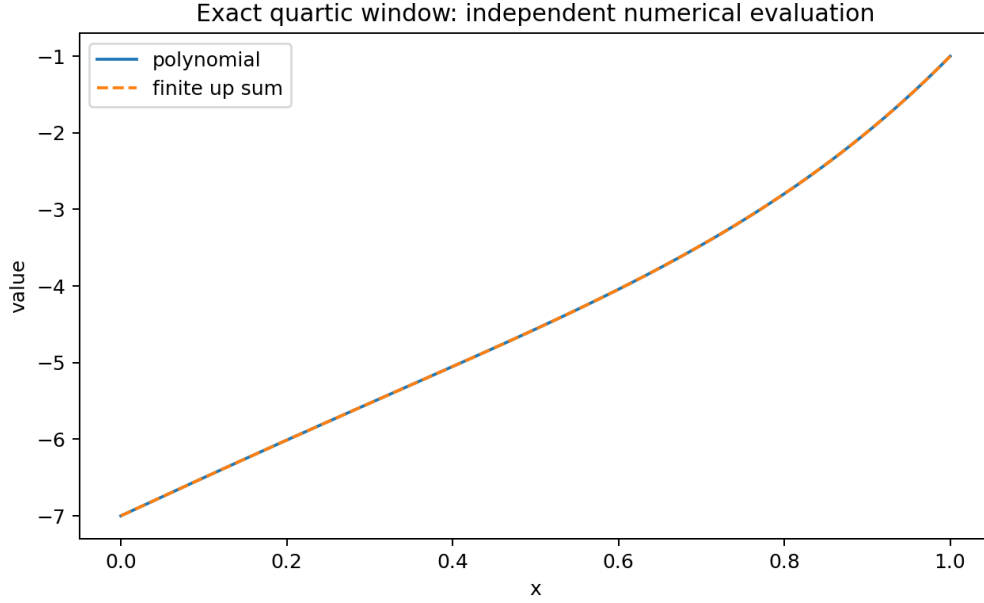


Figure 1: The quartic and its finite ten-atom Rvachev- u representation. The curves are visually indistinguishable.

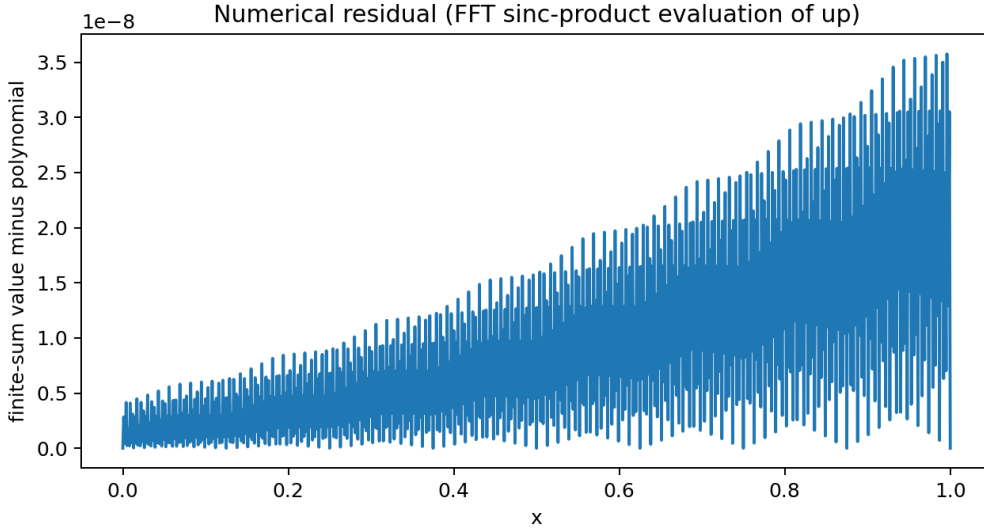


Figure 2: Residual from an independent FFT evaluation of the infinite sinc product. Exact symbolic verification gives zero.

13 Numerical conditioning

The sparse representation is algebraically simple but not automatically well-conditioned in floating-point arithmetic. Its level normalization is

$$n!2^{n(n+1)/2}, \quad (74)$$

which grows faster than exponentially. The paired atoms are sampled near increasingly flat portions of their large supports, so the polynomial emerges from cancellation among large terms. This is a familiar tension between exact compact smoothness and numerical localization.

Recommended practice is:

- construct b_n and $\alpha_{n,N}$ in exact rational arithmetic;
- evaluate with extended or arbitrary precision for moderate or high d ;
- sum the atoms in equal-amplitude blocks, ordered by degree;
- use compensated or pairwise summation;
- reduce h when a shorter support and smaller individual polynomial coefficients are worth the increased atom count;
- optimize h against an application-specific condition functional rather than choosing $h = L/2$ automatically.

The quasipolynomial estimate in [Theorem 4.6](#) explains why reducing h does not simply make every coefficient worse. As $K \sim L/(2h)$ grows, the normalized binary weights behave like N^n , while the Appell coordinates of a fixed polynomial carry powers of h . There is substantial balancing, and a nontrivial conditioning optimum should exist in many norms.

14 Additional deductions

14.1 A reflected anticausal formula

By symmetry, the right-tail operator

$$(J_+ f)(x) = \int_x^\infty f(t) dt \quad (75)$$

satisfies

$$J_+ u(x) = \sum_{k \geq 0} u\left(\frac{x + (2k + 1)}{2}\right). \quad (76)$$

Iterating gives a mirror of [Theorem 4.3](#). It yields exact polynomial windows whose long support tail lies to the left instead of the right. The two orientations can be mixed degree by degree because any sequence of degree- n monic polynomials remains a triangular basis. This introduces a discrete support-balancing design variable absent from the one-sided formula.

14.2 Analytic-function approximation

Let f be analytic in a neighborhood of $[a, b]$. Expanding $\tilde{f}(y) = f(c + hy)$ in the Appell basis and truncating at degree d gives an explicit finite u sum. Since $G(D)$ is an entire formal operator on polynomials, the coefficients are computable from derivatives of f at c . Any standard polynomial approximation remainder immediately becomes a remainder bound for finite u sums. Thus the exact polynomial theorem is also a direct constructive approximation scheme for analytic and sufficiently smooth data.

14.3 Tensor products

On a box $I_1 \times \cdots \times I_s$, tensor products of the univariate block basis represent every multivariate polynomial exactly:

$$\prod_{j=1}^s \Psi_{n_j}^{I_j}(x_j) \quad (77)$$

is a finite sum of tensor-product u atoms. A coordinatewise degree vector $(d_1, \dots, d_s)$ requires at most $\prod_j 2(d_j + 1)$ raw tensor atoms in the direct sparse construction. Low-rank and total-degree compression are open optimization questions.

14.4 Hermite jet stitching

Because (54) gives explicit endpoint derivatives, one can solve finite Hermite systems to build compactly supported C^∞ functions that agree with different prescribed polynomials on several disjoint intervals. The gaps carry smooth transitions automatically. This offers a new atomic-function analogue of polynomial splines with infinitely smooth knots outside the exact polynomial plateaus.

15 Conjectures and frontier problems

The following statements are deliberately separated from the proved results.

Conjecture 15.1 (Generic atom minimality in causal dyadic ladders *conjectural*). *Fix d and require a representation to use a single causal dyadic scale ladder, with every degree block having a degree-independent left support edge and a polynomial tail of exact degree n . For a Zariski-open set of degree- d polynomials, at least two atoms are necessary at every active level. Hence the canonical $2d + 2$ count is minimal in this class.*

The restriction to a precise causal class is essential. Global unrestricted minimality is not claimed; nonlinear choices of centers and scales may permit fewer atoms for special polynomials.

Conjecture 15.2 (Support-atom Pareto optimality *conjectural*). *For fixed degree d and collar size $\rho/L \rightarrow 0$, any causal positive-scale construction with uniformly controlled block condition number requires $\Omega_d(L/\rho)$ atoms. The bound in (64) is therefore optimal up to a degree-dependent constant.*

Conjecture 15.3 (Conditioning-optimal scale *conjectural*). *For every nonzero p and interval $[a, b]$, natural coefficient and evaluation condition functionals have at least one finite minimizer in $h > 0$. After factoring the integer jumps of $K = \lceil L/(2h) \rceil$, the minimizer is unique or belongs to a short finite set of adjacent K regimes.*

Conjecture 15.4 (q -ary refinable densities *conjectural*). *Let a compact density arise from a geometric random series with ratio q^{-1} , $q \in \mathbb{N}$, and an appropriate uniform digit law. Whenever its derivative refinement has an integral translation stencil, repeated antiderivatives admit weights generated by*

$$\prod_{j=0}^{m-1} (1 - z^{q^j})^{-1},$$

and exact finite polynomial windows follow by the same support truncation. The dyadic case is the most arithmetic because the reciprocal stencil is a Thue–Morse block.

Problem 15.5 (Unrestricted minimal atom number). *Let N_d be the smallest integer such that every real polynomial of degree at most d can be represented on every nondegenerate interval by at most N_d shifted and positively scaled u atoms with arbitrary real amplitudes. Determine N_d . This report proves*

$$N_d \leq 2d + 2, \quad N_0 = 2.$$

Even N_1 appears to be a useful nontrivial test problem.

Problem 15.6 (Symmetric support optimization). *Combine causal and anticausal blocks, allowing degree-dependent orientation and possibly unequal h_n , to minimize the maximum left/right collar subject to an atom budget and a condition-number constraint. Determine whether the exponential one-sided scale growth can be reduced without increasing the atom count above a linear function of d .*

Problem 15.7 (Thue–Morse and q -binomial expansions). *Expand the Green weights $p_m(N)$ and their normalized versions $w_{n,N}$ in bases adapted to Gaussian q -binomial coefficients, Prouhet cancellation, and Mahler functional equations. Seek identities that make the lower-order quasipolynomial coefficients explicit and connect them to Bernoulli polynomials on residue classes modulo 2^{m-1} .*

Problem 15.8 (Dual functionals and quadrature). *Construct compactly supported dual functionals λ_n satisfying $\lambda_n(\Psi_m) = \delta_{nm}$. Determine whether they can be expressed by finite dyadic samples or moments of u , and use them to derive exact quadrature or local projection formulas.*

Problem 15.9 (Formal verification). *A Lean formalization can be divided into four small modules:*

1. *the one-step telescoping identity;*
2. *the binary-partition induction;*
3. *the support-truncation theorem;*
4. *the finite-dimensional Appell operator inversion.*

None requires delicate asymptotic analysis, making the main theorem a promising formalization target for the repository.

16 Implementation and reproducibility

The archive accompanying this report contains

- `up_polynomial_representation.py`: exact construction, verification, and numerical sinc-product evaluation;
- `generated/quartic_atoms.csv`: the exact atom table;
- `generated/quartic_numerical_check.csv`: 1001-point numerical comparison;
- `generated/numerical_summary.txt`: exact coefficients and error statistics;
- the two figures used above.

A complete reproduction is obtained with

```
python up_polynomial_representation.py --output-dir generated
```

The symbolic identity is independent of numerical evaluation. The numerical code computes the Fourier coefficients

$$\frac{1}{P} \hat{u}\left(\frac{2\pi k}{P}\right)$$

on a period $P = 4 > 2$, truncates the infinite sinc product after 44 factors, and uses an inverse FFT. This is an independent check because the construction uses the refinement/antiderivative identity while the numerical evaluator uses the Fourier product.

17 Conclusion

The exact local polynomial-representation problem reduces to a surprisingly rigid chain of inversions:

refinement difference $\longrightarrow$ antiderivative train $\longrightarrow$ binary partitions $\longrightarrow$ Appell polynomial tail $\longrightarrow$ finite support truncation.

This chain yields an arbitrary-order algorithm with closed coefficients and no numerical solve. In its sparsest form, degree d costs only $2d + 2$ shifted and scaled copies of one fixed C^∞ compactly supported function. In its localized form, support collars can be prescribed at the cost of more atoms. The same formulas expose a direct bridge among the Rvachev refinement equation, the infinite sinc product, Bell–Bernoulli cumulants, restricted binary partitions, and the signed Thue–Morse sequence.

Classical Rvachev theory already guarantees that suitable atomic-function spaces contain polynomials. The value of the present construction is its explicitness: every center, scale, amplitude, support endpoint, determinant, and arithmetic coefficient is available in closed form and can be verified in exact rational arithmetic. This makes the theorem suitable both for further analysis and for formalization.

A Detailed proof of the coefficient identities

For completeness, we expand the operator calculation. From (22),

$$A_n(y) = \sum_{r=0}^n \binom{n}{r} \mu_r y^{n-r} = M(D)y^n.$$

Suppose $\tilde{p} = \sum_n b_n A_n$. Since $G(t)M(t) = 1$ as formal power series,

$$G(D)\tilde{p} = \sum_n b_n G(D)M(D)y^n = \sum_n b_n y^n.$$

Now

$$G(D)\tilde{p}(y) = \sum_{r=0}^d \gamma_r \frac{\tilde{p}^{(r)}(y)}{r!}.$$

Taking n derivatives and evaluating at zero gives

$$\begin{aligned} b_n &= \frac{1}{n!} \sum_{r=0}^{d-n} \frac{\gamma_r}{r!} \tilde{p}^{(n+r)}(0) \\ &= \sum_{r=0}^{d-n} \frac{\gamma_r h^{n+r}}{n! r!} p^{(n+r)}(c). \end{aligned}$$

If $\tilde{p} = \sum_j c_j y^j$, then

$$[y^n] \frac{D^r}{r!} \tilde{p} = \frac{(n+r)!}{n!r!} c_{n+r} = \binom{n+r}{r} c_{n+r},$$

which proves (45).

B Exact support bookkeeping

For the general atom

$$u\left(\frac{x - a - h(2^{n+1} - 2 + 2N)}{2^{n+1}h}\right),$$

the center and radius are

$$C_{n,N} = a + h(2^{n+1} - 2 + 2N), \quad H_n = 2^{n+1}h.$$

Hence

$$\begin{aligned} C_{n,N} - H_n &= a - 2h + 2hN, \\ C_{n,N} + H_n &= a + h(2^{n+2} - 2 + 2N). \end{aligned}$$

The minimum occurs at $N = 0$ and the maximum at $(n, N) = (d, K)$, proving (61). Since $K < L/(2h) + 1$,

$$\begin{aligned} a + h(2^{d+2} - 2 + 2K) &< a + h(2^{d+2} - 2) + L + 2h \\ &= b + 2^{d+2}h. \end{aligned}$$

C A compact table of exact formulas

Table 2: Formula sheet for implementation.

Object	Exact formula
Cumulant	$\kappa_{2m} = 2^{2m-1} B_{2m}/(m(2^{2m} - 1)), \kappa_{2m+1} = 0.$
Moment recurrence	$\mu_0 = 1, \mu_n = \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \mu_{n-r}.$
Reciprocal MGF	$G(t) = 1/M(t) = \sum \gamma_r t^r / r!.$
γ recurrence	$\gamma_0 = 1, \gamma_n = -\sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \gamma_{n-r}.$
Appell polynomial	$A_n(y) = \sum_{r=0}^n \binom{n}{r} \mu_r y^{n-r}.$
Binary weight	$p_m(N) = [z^N] \prod_{j=0}^{m-1} (1 - z^{2^j})^{-1}.$
Truncation index	$K = \lceil (b - a)/(2h) \rceil.$
Basis atom center	$C_{n,N} = a + h(2^{n+1} - 2 + 2N).$
Basis atom scale	$H_n = 2^{n+1}h.$
Basis block	$\Psi_{n,h} = n! 2^{n(n+1)/2} \sum_{N=0}^K p_{n+1}(N) u((x - C_{n,N})/H_n).$
Polynomial coordinate	$b_n = \sum_{r=0}^{d-n} \gamma_r h^{n+r} p^{(n+r)}(a - h)/(n!r!).$
Raw amplitude	$\alpha_{n,N} = n! 2^{n(n+1)/2} b_n p_{n+1}(N).$

Object	Exact formula
Support	$[a - 2h, a + h(2^{d+2} - 2 + 2K)]$.

D Repository comparison checklist

The following checks were used to avoid accidental duplication of the reviewed corpus:

1. Search the representation syntheses for “polynomial reproduction”, “Strang–Fix”, “finite polynomial”, “binary partition”, and equivalent variants.
2. Compare with the repeated-integration material, which studies convergence of finite box-convolution iterates toward u rather than inversion of the refinement derivative into a polynomial tail.
3. Compare with the existing Appell sections, which organize dyadic Taylor blocks of the Fabius function rather than interval-local sums of u atoms.
4. Compare with the sinc-product and Fourier-zero sections, which supply the MGF/Fourier data used here but not the support truncation.
5. Compare with the Thue–Morse and q -series drafts, where finite Prouhet products occur but not as the discrete Green operator for repeated antidifferentiation.
6. Check the historical/transcribed papers and, crucially, Rvachev’s 1990 survey for classical polynomial-inclusion claims.

The result is best viewed as a new synthesis and explicit constructive layer built from known ingredients, not as a claim that the broad phenomenon of polynomial representation by atomic functions was previously unknown.

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